

Project Design Phase Solution Architecture

Date	04 October 2026
Team ID	SWTID-2026-1268
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. For this project it explains how the ServiceNow platform decides, for each record, whether a user may see it, by running an ACL script that compares the record's field value with the user's roles. Its goals are to:

- Find the best tech solution to the problem of over-broad record access.
- Describe the structure, characteristics and behaviour of the solution to project stakeholders.
- Define features, development phases and solution requirements.
- Provide the specification according to which the solution is defined, managed and delivered.

Solution Architecture Diagram:

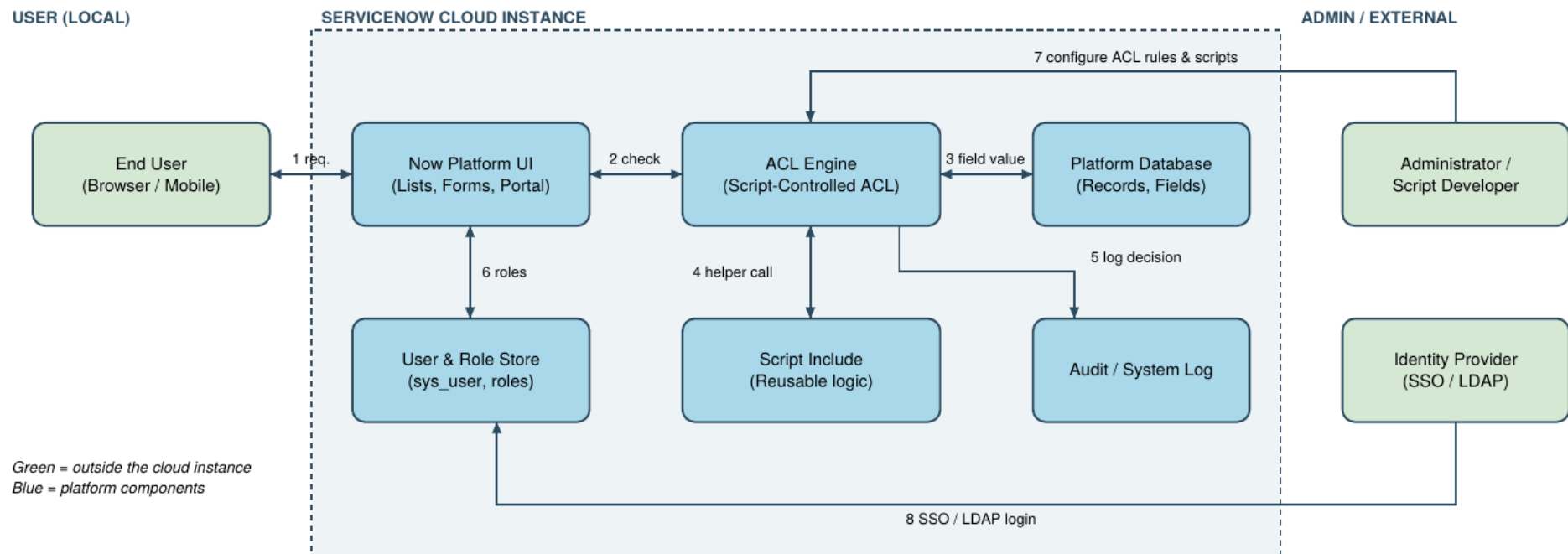


Figure 1: Architecture and data flow of the Script-Controlled ACL solution

Components

Component	Role in the solution	Technology
End User (Browser / Mobile)	Requests lists and records	Web browser, mobile app
Now Platform UI	Shows lists, forms and portal pages; passes each request to the ACL engine	ServiceNow Next Experience UI / Service Portal
ACL Engine	Runs the ACL rule and script for the request and returns allow or deny	ServiceNow ACLs, server-side JavaScript
Platform Database	Holds the records and the controlling field value (e.g. Department)	ServiceNow tables
User & Role Store	Provides the user's roles and attributes	sys_user, role tables
Script Include	Reusable helper logic shared by several ACL scripts	JavaScript Script Include
Audit / System Log	Records every allowed and denied attempt	ServiceNow logs
Administrator / Script Developer	Creates and tests ACL rules and scripts	Studio, ACL configuration
Identity Provider	Authenticates users through single sign-on	SAML 2.0 SSO, LDAP

Data Flow (numbers match the diagram):

1. The user requests a list or record from the Now Platform UI.
2. The UI passes the request to the ACL engine for a check.
3. The ACL script reads the record's field value from the platform database.
4. The script calls shared helper functions in a script include when needed.
5. The allow or deny decision is written to the audit log.
6. The user's roles and attributes are looked up in the user and role store.
7. Administrators configure ACL rules and scripts.
8. Users sign in through SSO / LDAP.

Key design points:

- Access is denied by default; script errors and empty fields deny access.
- The check is enforced on the server for forms, lists, reports and API calls, so it cannot be bypassed from the browser.
- Cheap conditions run before scripts and condition fields are indexed to keep response time low.
- The cloud platform provides availability and lets the solution scale to more users, tables and records.